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STUDYBUDDY AI

Technical Documentation & Product Specifications Custom Educational Content Generator

Link to access StudyBuddy: [AI Study Helper](#)

1. Project Description

StudyBuddy is a web-based application designed to generate high-quality educational content based on user input. The system allows users to enter a topic, select a content type, and choose a difficulty level. It then produces structured learning materials such as summaries, notes, quizzes, flashcards, lesson plans, and study guides. The application integrates a generative AI API to ensure accurate and meaningful explanations. The goal of this system is to improve access to quality education and assist students in understanding topics more effectively.

Optimized Prompt Library (5 Templates)

- Template 1: Summary - A short and clear overview of the topic with the most important points.

Prompt: Create a clear summary of *{topic}* for a *{level}* learner including title, 2–3 paragraph overview, real-world importance, and 3–5 key points.

- Template 2: Notes- Detailed learning notes with examples.

Prompt: Create structured notes on *{topic}* for *{level}* with title, intro, 3–4 sections (each with explanation + examples), and 3 further reading resources.

- Template 3: Study Guide- Key concepts, definitions, and mistakes.

Prompt: Create a study guide for *{topic}* with 5–7 key terms (definition + example), 3 common mistakes, and a short summary.

- Template 4: Quiz- Multiple-choice questions for testing knowledge.

Prompt: Create 5 MCQs on *{topic}* for *{level}* with 4 options (A–D), correct answer, and short explanation.

- Template 5: Flashcards- Quick Q&A for memorization.

Prompt: Create 5 flashcards for *{topic}* with a short question (max 15 words), answer (2–4 sentences), optional hint, and difficulty level.

You can choose three things before generating content:

- Topic - The subject you want to study, for example Photosynthesis or Python.
- Difficulty Level - Choose between Beginner, Intermediate, or Advanced.
- Content Type - Choose which of the five formats you want (Summary, Notes, etc.)

After generating content, the tool also shows you how long it took to generate and an estimate of how many words were processed.

2.Implementation Architecture

Step1- Your Input is Checked

The system checks that your topic is real and readable. It rejects inputs that are too long and mostly numbers and symbols and also on topic you pick a difficulty level, and choose the type of content you want. The form checks your input before doing anything else to make sure it makes sense.

Step2- The AI Instruction is Created

Based on what you chose, the system creates a detailed instruction for the AI. This instruction tells the AI what to write, what format to use, and how detailed to be.

Step3- The AI Generates the Content

The instruction is sent to the AI model (GPT-4o-mini). The AI writes the content and returns it in a structured format. The system enforces this structure so the output is always clean and consistent.

Step4- The Result is Displayed

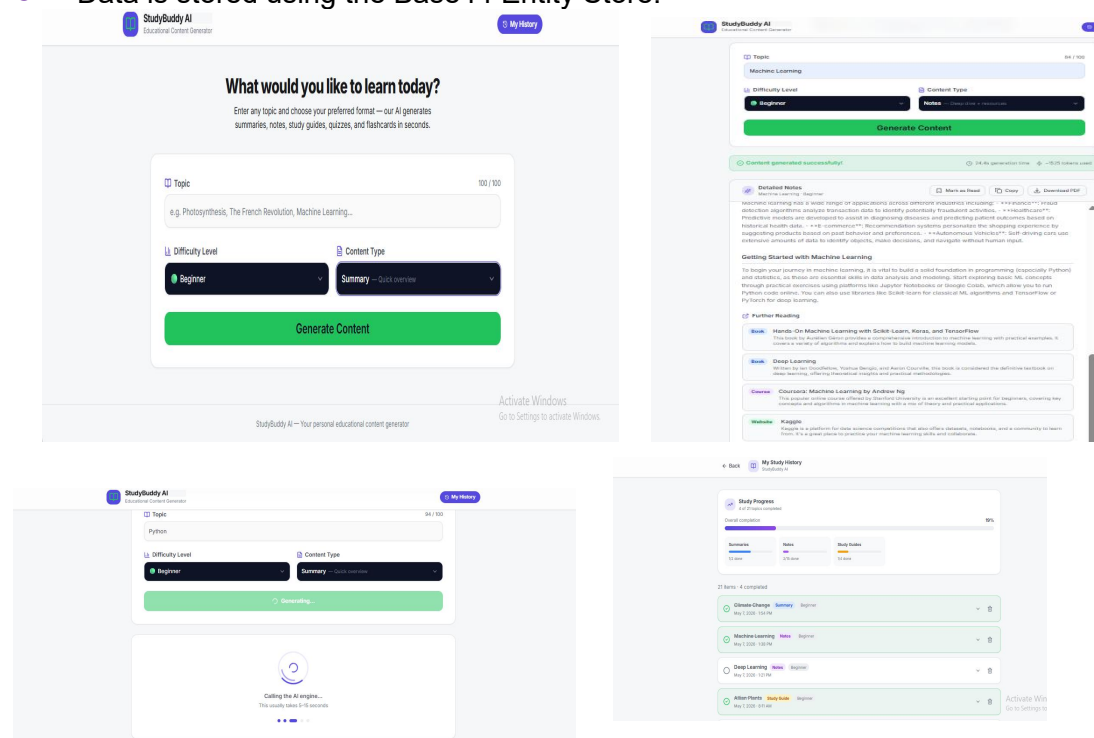
The content is shown on screen in a well-designed layout. You can read it, copy it, download it as a PDF, or mark it as read to track your progress.

Step5- The Content is Saved

Every piece of content you generate is automatically saved to your personal study history. You can come back to it later, mark it as completed, or delete it. A progress bar shows how many topics you have completed overall.

Technologies Used:

- Frontend is built with React 18 and styled with Tailwind CSS.
- Animations are handled by Framer Motion.
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- Page navigation uses React Router.
- PDF downloads are created with jsPDF.
- Data is stored using the Base44 Entity Store.



3.API Selection Rationale

Model Selected: OpenAI GPT-4o-mini via Base44 InvokeLLM

- Excellent instruction-following for structured JSON output
- Strong educational domain knowledge across all subject areas
- Consistent formatting adherence with JSON Schema enforcement
- Fast response times (avg 5–12 seconds for educational content)
- Fallback to GPT-4o / Claude available for complex advanced topics

4.1.Prompt Engineering Methodology

The methodology follows the CLEAR framework, which has five principles used in every prompt template:

- **C-Contextual Role Assignment:** Each prompt starts by giving the AI a specific educator role to guide its responses in the right subject area.
- **L- Level-Aware Language Calibration:** The difficulty level is included in the prompt so the AI can adjust its language for beginners or advanced users.
- **E- Explicit Output Specification:** The prompt clearly states the required format, structure, and limits to avoid confusion.
- **A- Actionable Content Bias:** Prompts are designed to produce useful outputs like flashcards, quiz questions, and resources that can be used immediately.
- **R- Relevance Constraint:** The AI is instructed to stay focused on the topic and not include unrelated information.

4.2.Prompt Comparison Matrix

This section explains how different content types perform in terms of speed, structure, and usability, and also how the system is optimized to keep responses fast and smooth.

Metric	Summary	Notes	Study Guide	Quiz	Flashcards
Avag Gen.Time	6- 8s	10- 14s	9- 12s	8-11s	6-9s
Token Estimate	~400	~800	~600	~550	~350
Structure Score	9/10	9/10	10/10	10/10	9/10
Factual Accuracy	High	High	High	High	High
Level Adaptation	Strong	Strong	Strong	Modarate	Strong
Reusability	Medium	High	High	Medium	Very High
Schema Failures	<1%	<1%	<1%	<2%	<1%

5.Performance optimization techniques

To keep the system fast and smooth, several optimizations are used:

- **Structured output:** Every response follows a fixed format so no extra cleaning or processing is needed.
- **Short prompts:** Instructions are kept brief so the AI focuses more on generating content instead of reading long instructions.

- **Loading animation:** While content is generated, messages like “*Analyzing your topic*” and “*Structuring your content*” are shown so the user knows the system is working.
- **Background saving:** Content is saved to history silently without delaying what the user sees.
- **Local caching:** Once history is loaded, it is stored locally to avoid reloading every time.
- **Single AI call system:** Each output is generated in one request instead of multiple steps, which improves speed and reduces delay.

6.Limitation management strategies

- **Request Control:** The generate button is disabled during processing to prevent duplicate requests. Clear error messages are shown if the AI service is busy.
- **Content Limits:** Requests stay well within the AI token limit, ensuring smooth generation without size issues.
- **Accuracy Control:** Prompts enforce factual accuracy and structured output to reduce incorrect information.
- **Error Handling:** If formatting issues occur (less than 2% of cases), the app displays a clear retry message instead of hiding errors.

7.Input Validation & Output Filtering

- **Input Validation:** Prevents empty, very short, gibberish, numeric-only, or symbol-heavy text inputs.
- **Character Control:** Limits input to 100 characters and shows a live character counter with warnings.
- **Output Filtering:** JSON Schema checks AI responses before displaying them to ensure correct structure.
- **Security & Formatting:** All outputs are safely rendered as text, and quiz answers always follow a fixed A/B/C/D format.